

Weekly Project Report –Week 5 Unified Generative AI Models via CommandLine Interface

Group 6 Section: 05

Araf Tahsan Pavel ID: 2212079042

Md Tanveer Hossain Nihal ID: 2211935042

Ismot Ara Emu ID: 2212108042

Objective

Extend our unified CLI to support both vision-based generative models (DCGAN, CGAN, CVAE) **and** a text-generation model (GPT-2) while ensuring the codebase remains modular and conflict-free.

Last Week Week 4)

Conditional Variational Autoencoder (CVAE)

- Designed encoder/decoder networks with label embeddings.
- Applied the re-parameterization trick for latent sampling.
- Combined Binary Cross-Entropy reconstruction loss with KL-divergence, using a β -weighting schedule.
- Trained the CVAE with Adam, visualising class-conditioned reconstructions.

Week 5

☐ **Integrated GPT-2 fine-tuning**

- Added codeP/models/gpt2.py with the full fine-tuning and generation class.
- Wrote codeB/gpt2.sh so one command handles train + inference.
- Extended codeP/runner.py to recognise --model gpt2.

☐ **Dataset preprocessing**

- Created project_datasets/gpt2_dataset.py to ingest PDF/DOCX/TXT, clean text and build TextDataset objects.
- Implemented cache handling and configurable block-size.

☐ **Namespace conflict fix**

- Tracked down a clash between our local datasets/ package and the Hugging Face datasets library.
- Added a five-line shim at the top of gpt2.py; the local module is now aliased as local_datasets, allowing the external library to load correctly.